Cristhian Victor Rojas Marquez

Software Engineer | Algorithmic Problem Solver & AI-Native Developer

Cochabamba, Bolivia | +591 68301052 | cristhiabnvictorrm@gmail.com
linkedin.com/in/rojasredo | github.com/RedoRojo | redoportfolio.vercel.app

Professional Summary

Systems Engineering Honors graduate and passionate competitive programmer (**3x ICPC National Finalist in 2023, 2024, and 2025**) driven by complex algorithmic problem-solving and analytical thinking. Deeply enthusiastic about the future of **Artificial Intelligence as a transformative force in software engineering**—focusing on high-level system design, critical reasoning, and orchestrating **AI-native and agentic workflows** to solve hard engineering problems beyond routine code generation.

Technical Skills

Programming Languages: TypeScript, JavaScript, Python, C++, Java, SQL

Frameworks & Databases: React, Next.js, NestJS, FastAPI, Node.js, Tailwind CSS, PostgreSQL, SQLite, SQLAlchemy, TypeORM

AI & Agentic Workflows: Claude Code, OpenAI Codex, Cursor, GitHub Copilot, Google GenAI SDK (Gemini API), LLMs

Core Engineering: Data Structures & Algorithms, RESTful APIs, OOP, Unit & Dynamic Testing, Debugging, Git, Docker, Linux

Languages: Spanish (Native), English (Advanced – Professional Working Proficiency)

Education

Universidad Católica Boliviana "San Pablo" **Feb 2021 – Feb 2026**
Bachelor of Science in Systems Engineering *Cochabamba, Bolivia*

- **Graduated with Honors:** Awarded *Graduation by Excellence* (Mar 2026) in recognition of outstanding academic performance.
- **Relevant Focus:** Data Structures & Algorithms, Software Architecture, Object-Oriented Programming, Automated Testing, Databases, and Artificial Intelligence.

Experience

University Lecturer – Data Structures **Aug 2026 – Present**
Universidad Católica Boliviana "San Pablo" *Cochabamba, Bolivia*

- Lead undergraduate instruction for the **Data Structures** course, designing lectures, algorithmic evaluations, and coding labs focused on time/space complexity, trees, graphs, and memory-efficient design.
- Previously served 6 semesters (Feb 2023 – Dec 2025) as Teaching Assistant for **Data Structures**, **Programming I (OOP)**, and **Introduction to Programming**, conducting lab sessions and code reviews.

Independent AI Engineer Consultant **Aug 2026 – Sep 2026**
Video Game Venture & Investment Fund (Independent Contract) *Remote*

- Engineered and maintained an AI-powered investment decision-support system that processes and synthesizes multi-source gaming market intelligence through **Large Language Models (LLMs)** to guide investor evaluations.
- Leveraged agentic coding workflows (**Claude Code** and **Codex**) across the entire development and maintenance lifecycle to rapidly ship data processing pipelines and analytical modules.

Research Assistant – Software Testing & Dynamic Analysis **Aug 2025 – Dec 2025**
Pontificia Universidad Católica de Chile *Remote*

- Developed a **Python**-based dynamic code analysis tool to evaluate software reliability and automated test suite efficacy.
- Automated runtime execution tracing and metric extraction to detect silent assertion mismatches where test assumptions diverged from actual program behavior.

Full-Stack Software Developer (Freelance) **Jul 2025 – Nov 2025**
Enterprise Administration System *Cochabamba, Bolivia*

- Collaborated within an agile team to build a full-stack enterprise administration platform using **React**, **NestJS**, and **TypeScript**.
- Designed and consumed modular **REST APIs** applying strict **OOP** principles, implementing validation layers and relational data persistence.

Technical Projects

Chasqui – Longitudinal News & Blindspot Intelligence | *FastAPI, Next.js 15, pgvector, Docker* **GitHub Repository**

- Built a full-stack news intelligence platform ingesting daily reports from **46 Bolivian outlets** across all 9 departments using concurrent RSS/DOM scrapers, NLP geo-tagging, and GitHub Actions CI.
- Engineered a two-stage semantic clustering pipeline (`Articles` → `Events` → `Story Threads`) backed by 128d embeddings in **PostgreSQL + pgvector** to track evolving timelines and quantify media blindspots.

Librerías Bolivia – Book Price Aggregator & API | *FastAPI, Next.js, TypeScript, PostgreSQL* **GitHub Repository**

- Built a full-stack platform and **RESTful API** consolidating real-time book pricing across **14 Bolivian bookstores and publishers** via **Shopify** and **WooCommerce REST APIs** with a 4-level cascade deduplication engine.
- Implemented a responsive **Next.js (React, TypeScript)** frontend and accent-insensitive fuzzy search in **PostgreSQL** (`pg_trgm`, `unaccent`), backed by automated unit tests.

Escriba – AI Spanish Linguistic Analyzer | *Python, FastAPI, Gemini 2.5 SDK, Tailwind CSS* **GitHub Repository**

- Developed an AI-native web tool integrating the **Google GenAI SDK (Gemini 2.5 Flash)** via a **FastAPI** REST backend to classify errors across spelling, grammar, punctuation, semantics, and style using structured JSON schemas (*arXiv:2511.01615*).

Volunteering & Leadership

Deep Learning Tutor (Training Team) | *Bolivian Delegation – IOAI* **Jul 2026**

- Supported the preparation of Bolivia's national team for the **International Olympiad in Artificial Intelligence (IOAI)** as part of the coaching team, teaching **Deep Learning** foundations, neural network architectures, and practical model training.

President | *Digital Brain – Neurotechnology Student Society, UCB Cochabamba* **Jan 2025 – Jan 2026**

- Led the university's neurotechnology student society, organizing interdisciplinary study groups, technical workshops, and student research activities.

President | *Sociedad Científica de Ingeniería de Sistemas – SCEIS UCB* **Feb 2024 – Dec 2025**

- Directed organizational leadership, student activities, and academic engineering events across the Systems Engineering department.

Competitive Programming Tutor & Contest Organizer (ICPC) | *UCB Cochabamba* **Feb 2023 – Dec 2025**

- Mentored 20+ engineering students in competitive programming fundamentals, algorithms, and systematic debugging.
- Planned and organized 4 internal university programming contests, managing registration, problem preparation, and automated test suites.

Competitive Programming Honors

- **3x ICPC National Finalist (2023, 2024, 2025):** Qualified consecutively across three seasons to represent the university at the national finals.
- **4th Place Nationwide – 2025 ICPC South America South Finals (Nov 2025):** Ranked among the top competitive programming teams in Bolivia solving complex algorithmic problems under strict time constraints.
- **Silver Medal – "SanSi Cup 2023" Programming Contest (Dec 2023)** & **7th Place Nationwide – 2023 ICPC Bolivia Preliminary Contest (Sep 2023)**.